Alvin Ghindăoanu.

Industrial Engineering student

Bucharest, Romania alvinmarc17@yahoo.com

PROFILE

Final-year Industrial Engineering student at POLITEHNICA Bucharest, back from an Erasmus year at UPM Madrid. I design in CATIA, SolidWorks and AutoCAD, and I am looking for an internship or an entry-level engineering role.

EDUCATION

Bachelor's degree — Industrial Engineering 2023 – 2027
POLITEHNICA Bucharest — Faculty of Industrial Engineering and Robotics · Final year

Erasmus+ exchange — full academic year 2025 – 2026
Universidad Politécnica de Madrid (UPM) — ETSIDI
Project-based courses in mechatronics systems design, IoT with Raspberry Pi and photovoltaic-powered products.

Bachelor's degree — Business Administration (taught in foreign languages) 2023 – 2027
Bucharest University of Economic Studies (ASE) — FABIZ
Studied in parallel with engineering; paused for one year during the exchange.

Baccalaureate — Mathematics & Computer Science, intensive English 2023
I.L. Caragiale National College, Bucharest

PROJECTS

PipeBot — pipeline inspection drone · UPM Madrid · Mechatronics Systems Design · Team of 4 2025–26
Compact ducted-fan quadcopter for inspecting pipes and tanks, designed in CATIA; the frame doubles as a collision cage around the sensors and flight electronics. **My part:** Co-designed the mechanical structure in CATIA and wrote the discussion, conclusions and cost analysis.

Cover part — CNC milling program · POLITEHNICA Bucharest · EdgeCAM lab · Individual
2.5D machining program in EdgeCAM for a 65 × 60 × 27 mm cover on a vertical mill: eight operations with end mills from 32 mm to 4 mm and three chamfering passes.

Roller chain — parts and assembly · POLITEHNICA Bucharest · SolidWorks · Individual
Roller chain modelled in SolidWorks: five parts (link plates, bushing, ring, pin) mated into a multi-link assembly.

Smart solar microgrid with wildfire early warning · UPM Madrid · IoT and Raspberry Pi · Team of 2 2025–26
Off-grid forest station that detects early signs of fire and raises the alarm over LoRa radio; solar, battery and supercapacitor system sized and checked with a 24-hour simulation. **My part:** Sized the solar, battery and supercapacitor system, built the 24-hour simulation and set up the LoRa radio link and its code.

PV-powered key tracker · UPM Madrid · Photovoltaic Powered Products · Team of 5 2026
Concept and energy budget for a key finder that never needs charging: indoor-light solar cell, Bluetooth and assisted GPS in a 40 mm disc. **My part:** Did most of the work: concept and user analysis, energy consumption profile, solar cell and battery sizing, and the physical layout.

Smart medicine box · UPM Madrid · Practical IoT with Raspberry Pi · Team of 3 2025
Working Raspberry Pi prototype that reminds a patient to take medication, confirms the box was opened and alerts a caregiver over MQTT. **My part:** Built most of the prototype: breadboard circuit, Python alarm logic, MQTT alerts and the web page.

SKILLS

CAD & CAM	SolidWorks, CATIA, AutoCAD, Fusion 360, EdgeCAM
Software & prototyping	Python (Raspberry Pi projects), Raspberry Pi, Excel, PowerPoint, Word
Strengths	Discipline, Determination, Composure under pressure, Working in international teams

ADDITIONAL

Languages	Romanian (Native) · English (C1–C2) · Spanish (B2) · German (A2)
Beyond engineering	Shotokan karate — World and European champion. More than 13 years of competition and still active. Certified national-level kumite referee. Volunteer — Chances for the Future (Șanse pentru viitor), 2022.
Other	Licences: Driving B · Motorcycle A2 · Interests: Snowboarding, Wakeboarding, Basketball